



Original Article

Linearity improvement of UltraScale+ FPGA-based time-to-digital converter



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ARTICLE INFO

Article history:

Received 3 May 2022

Received in revised form

16 September 2022

Accepted 11 October 2022

Available online 14 October 2022

Keywords:

Time-to-digital converter

Tapped-delay line

Field-programmable gate array

Histogram uniformization

Multi-chain TDL

ABSTRACT

Time-to-digital converters (TDCs) based on the tapped delay line (TDL) architecture have been widely used in various applications requiring a precise time measurement. However, the poor uniformity of the propagation delays in the TDL implemented on FPGA leads to bubble error and large nonlinearity of the TDC. The purpose of this study was to develop an advanced TDC architecture capable of minimizing the bubble errors and improving the linearity. To remove the bubble errors, the decimated delay line (DDL) architecture was implemented on the UltraScale + FPGA; meanwhile, to improve the linearity of the TDC, a histogram uniformization (HU) and multi-chain TDL (MCT) methods were developed and implemented on the FPGA. The integral nonlinearities (INLs) and differential nonlinearities (DNLs) of the plain TDCs with the 'HU method' (HU TDC) and with 'both HU and MCT methods' (HU-MCT TDC) were measured and compared to those of the TDC with 'DDL alone' (plain TDC). The linearity of HU-MCT TDC were superior to those of the plain TDC and HU TDC. The experiment results indicated that HU-MCT TDC developed in this study was useful for improving the linearity of the TDC, which allowed for high timing resolution to be achieved.

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1. Introduction

Time-to-digital converters (TDCs) have been widely employed in various applications that require precise time measurement such as positron emission tomography (PET) [1–5], time-resolved fluorescence spectroscopy [6,7], and light detection and ranging (LIDAR) system [8,9]. While the TDCs have been typically implemented on either application-specific integrated circuit (ASIC) or field-programmable gate array (FPGA), researches on FPGA-based TDC have been conducted more extensively in recent years due to its reconfiguration capability and lower development cost compared to ASIC-based TDCs [10–13].

With the advancements that have been made in FPGA manufacturing process technology, high-performance FPGA series such as UltraScale and UltraScale + FPGAs (Xilinx) have been newly introduced. These latest FPGAs provide smaller delay elements and

faster carry chains, which are favorable for designing TDCs with high time resolution based on the tapped delay line (TDL) architecture. However, the poor uniformity of the propagation delays in the TDL implemented on those FPGAs leads to bubble errors and large nonlinearity, both of which can degrade the precision of the TDC. To remove the bubble errors in the TDC, a bin realignment method that realigns the taps of the TDL according to their real delay time has been proposed [14]. However, this method required an additional repetition process to measure the real delay time between each tap in every TDC channel. Further, a few bubble errors still remained in the TDL after applying the bin realignment method. A bin decimation method that could make the tap delays as uniform as possible has been proposed to improve the linearity of the TDC [15]. However, this method required an additional offline process to calculate the bin width of the TDL.

The purpose of this study was to develop an advanced TDC architecture capable of minimizing the bubble errors and improving the linearity without the need for an additional repetition process and offline process. The decimated delay line (DDL) architecture was implemented on the FPGA to remove the bubble errors [16]. The multi-chain TDL (MCT) architecture was also implemented to improve the precision of the TDC [17]. The histogram uniformization (HU) method was newly developed in this study to improve

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the uniformity of the bin widths and combined it with the MCT method to improve the linearity. The linearity and precision of the TDC were measured to evaluate the performance of the advanced TDC developed in this study.

2. Materials & methods

2.1. Architecture of plain TDC with the decimated delay line architecture

TDC consists of a coarse time counter running at the system clock and time interpolator. In this study, the tapped delay line (TDL) was selected as a time interpolator generating a sub-clock fine timestamp because it had the simplest structure with efficient resource usage and easy encoding capability [18–20]. As shown in Fig. 1, the TDL was implemented using cascaded carry chain modules (CARRY8) in the UltraScale + FPGA; this was done because the total delay of the carry logic is much smaller in CARRY8 than CARRY4 in Xilinx FPGAs up to 7 series, which makes the delay between taps much finer. TDL-based TDCs implemented on FPGAs with CARRY8 can achieve a higher time resolution than those implemented on FPGAs with CARRY4. The delay length of the TDL was long enough (2.6 ns) to cover the clock period (2.5 ns).

When a signal was sent to the FPGA, a hit signal was generated by a pulse launcher and propagated along the TDL. The propagation states along the TDL were sampled by the D flip-flops, yielding a corresponding thermometer code. To remove the bubble errors that frequently appear in the thermometer code due to the clock skew and uneven propagation delays within the TDL, the decimated delay line (DDL) architecture that divides the TDL to sixteen sub-TDLs was implemented. The thermometer-to-binary encoder counted the consecutive '1's in the bubble-free thermometer code which indicates the position of the propagated signal, then generated a corresponding binary code. The binary code was converted to the fine time and the converted fine time was stored in a pre-built look up table (LUT). The coarse time was generated by multiplying the clock period (T) and the output of the counter (C) running at the clock frequency. Lastly, the final timestamp (T_F) was calculated by subtracting the fine time (T_{ft}) from the coarse time as [11].

$$T_F = (T \times C) - T_{ft} \quad (1)$$

One of the major problems of TDL-based TDC is the large variation in bin width, which leads to the generation of an inaccurate fine timestamp. This study used online bin-by-bin calibration to overcome this problem [21]. To perform bin-by-bin calibration, it is necessary to calculate a bin width, which is known as the code density test [22]. For this purpose, a hit signal uncorrelated with the system clock was generated and sent to the TDC. The propagation states along the TDL were sampled by the D flip-flops and a thermometer code was generated. The consecutive '1's in the thermometer code were counted by the thermometer-to-binary encoder and then the binary code corresponding to the bin number was generated. After this process was repeated with a large amount of hit signals, the histogram consisting of a bin number and its accumulated hit counts (bin width) was generated. The width (W_i) and fine time (T_{ft}) of the i -th bin with N_i hits were calculated as [23].

$$W_i = \frac{N_i}{N} \times T \quad (2)$$

$$T_{ft} = \sum_{k=0}^{i-1} W_k + \frac{W_i}{2} \quad (3)$$

where N and T are the total number of hit counts and the period of the clock, respectively. The bin numbers and calculated fine times were stored in the calibration look-up-table (LUT).

The delays in the carry logic vary depending on the fluctuation of the operating voltage and the temperature of the FPGA [24], which affects the bin width. There have been several approaches proposed to solve this problem, such as monitoring the frequency of a ring-oscillator [19] and continuously updating the calibration table for the bin width calculated by the code density test [25]. In this study, a ring oscillator based on NOT gates was implemented on the TDL. The frequency of the ring oscillator was measured and compared with the initial measurement in real time. Increased delays in the logic will lead to decreased frequency of the oscillator and vice versa. The current i -th bin width ($W_{cu,i}$) was calculated as

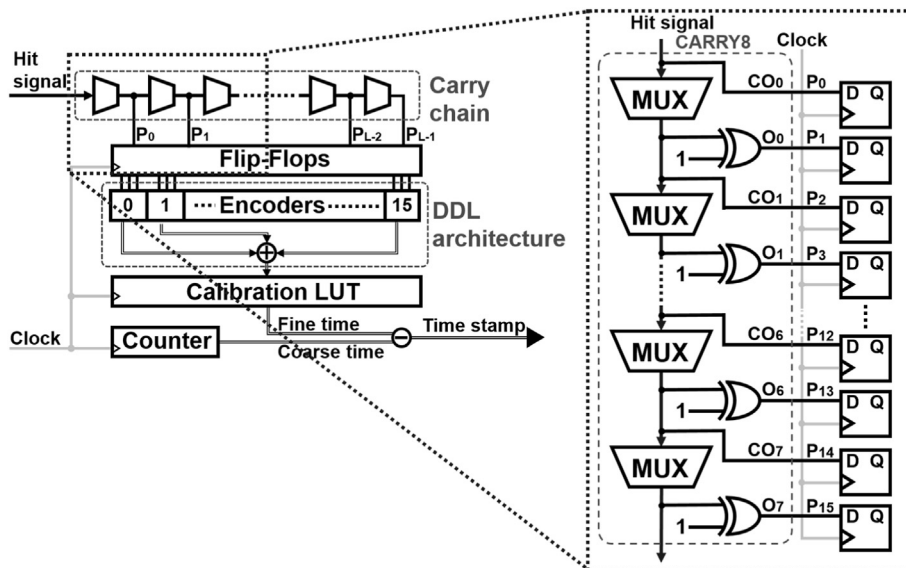


Fig. 1. The architecture of the plain TDC with the decimated delay line (DDL) implemented on the UltraScale + FPGA with the carry chain modules (CARRY8).

$$W_{cu,i} = \frac{f_{in}}{f_{cu}} \times W_{in,i} \quad (4)$$

where $W_{in,i}$, f_{cu} , and f_{in} , are the initial i -th bin width, current frequency, and initial frequency, respectively. If the bin width was changed, the values in the calibration LUT were updated in real time while the TDC was operating.

2.2. Architecture of advanced TDC with histogram uniformization method and the multi-chain TDL

2.2.1. Histogram uniformization method

One of the main problems of the TDL implemented in the UltraScale + FPGA was that the delays between neighboring taps were inconsistent with each other, thus resulting in poor uniformity of the bin widths. The poor uniformity of the bin widths degrades the linearity of the TDC, and consequently deteriorates the precision of the time measurement. To improve the uniformity of the bin widths and reduce the relative DNL and INL, we developed a histogram uniformization (HU) method that merges consecutive bins until their accumulated-bin width exceeds a predefined value, as shown in Fig. 2. The HU method was employed in plain TDC with a decimated delay line architecture.

To merge the bin widths, the numbers of hit counts that were stored at the original histogram generated by the bin-by-bin calibration were accumulated until exceeding a predefined value. The new histogram consisting of accumulated hit counts (bin width) and the merged bin number was generated. A mapping LUT was also generated to translate the original histogram to a new one. The bin numbers and fine times calculated using the new histogram were finally stored in the calibration LUT.

2.2.2. Multi-chain TDL

The limitation of the HU method is that wide bins can still remain after applying the HU method, which improves the linearity of the TDC. The effect of the wide bins can be alleviated by increasing the value of the threshold used in the HU method. However, this method causes an increase in the average bin width and a decrease in time resolution. To improve the precision of the TDC and reduce the absolute DNL and INL, various approaches have been proposed, such as combining multiple parallel TDLs into a single TDL [26,27], measuring low-frequency TDL multiple times [28], and sub-interpolating a TDL [29]. In this study, we implemented the multi-chain TDL method that combines multiple parallel TDLs into a single TDL, and we combined this method with the HU method.

As shown in Fig. 3, when a hit signal arrives at the TDC, the triggered signal simultaneously propagates through the parallel chains. The taps from each of the individual chains were collected and sampled by the D flip-flops. The total number of taps was increased by the number (J) of chains constructed in the multi-chain TDL. Compared to the single chain TDL, the bin widths were divided into smaller fractions according to the number of carry chains, which improved the precision of the TDC.

The overall architecture of the TDC employed with the DDL architecture, HU method, and the multi-chain TDL is shown in Fig. 4.

2.3. TDC implementation

Three types of TDCs were implemented on the UltraScale + FPGA (XCKU5P–2FFVB676E, Xilinx) in the KCU116 evaluation board: TDC having a DDL architecture alone (plain TDC), that with the HU method (HU TDC), and that with both the HU method and multi-chain TDL (HU-MCT TDC). The frequency of the system clock was set to 400 MHz because a higher clock frequency implies fewer bins in the delay line and less logic resources but also increases the probability of timing violation. The measurement range was up to 41.9 ms with a dead time of 7.5 ns. In total, 85 CARRY8s were cascaded to form a single TDL, and the TDL had a delay time slightly larger than the clock period of 2.5 ns. The operating temperature of the FPGA was maintained at 34 ± 0.5 °C through air circulation.

2.4. Performance evaluation

The differential nonlinearity (DNL) and integral nonlinearity (INL) were measured to evaluate the linearity of the plain TDC, HU TDC, and HU-MCT TDC. The relative DNL and INL were respectively defined as

$$Relative\ DNL_i = \frac{W_i - W_a}{W_a} \quad (5)$$

$$Relative\ INL_i = \sum_{k=0}^i DNL_k \quad (6)$$

where W_i and W_a are the i -th bin width and average bin width, respectively. The absolute DNL and INL were also calculated by multiplying the relative DNL and INL by the least-significant bit (LSB). To evaluate the effect of the chain numbers of the multi-chain TDL on the linearity, the number of chains was varied from two to four and the DNL and INL of HU-MCT TDC with each chain TDL were

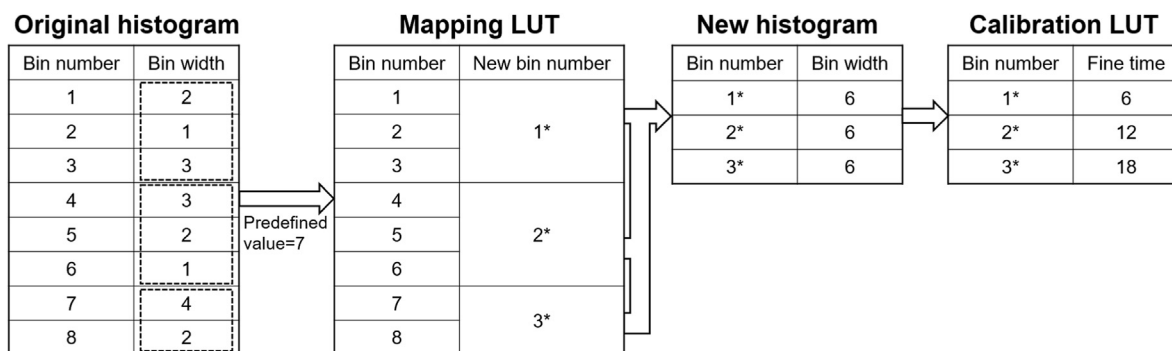


Fig. 2. New histogram and calibration look-up-table (LUT) generated from original histogram and mapping LUT using the histogram uniformization method.

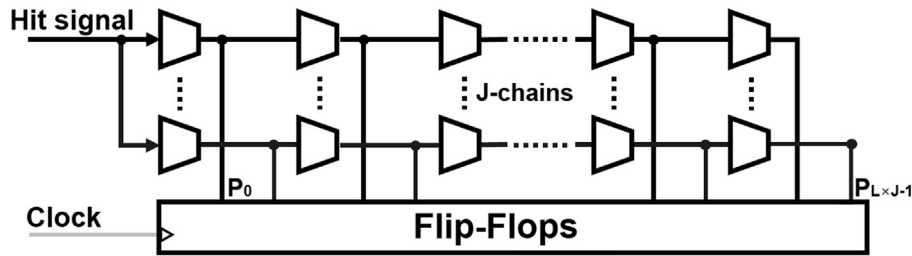


Fig. 3. The architecture of the multi-chain TDL wherein multiple (J) single chain TDLs were combined in parallel.

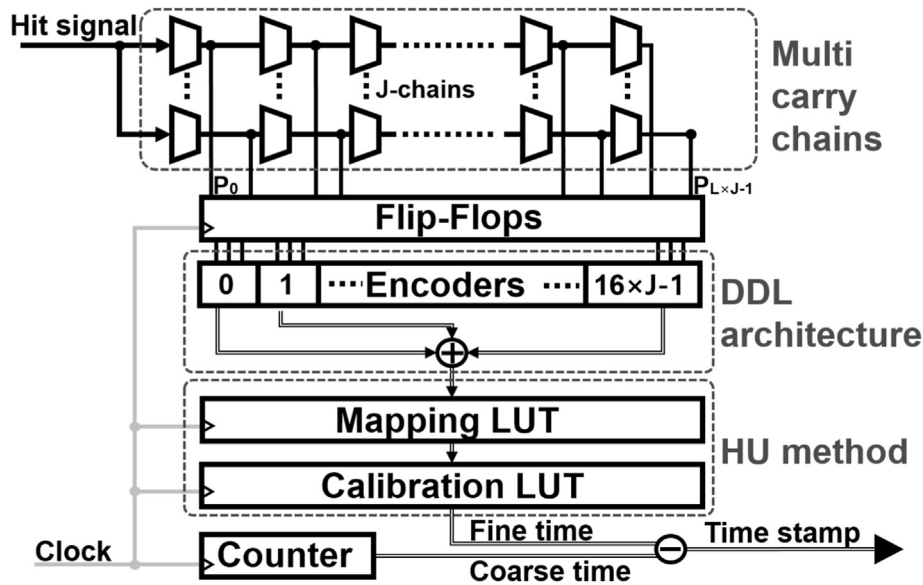


Fig. 4. The architecture of the TDC with the DDL architecture, the HU method, and the multi-chain TDL.

measured. In total, 1048576 (2^{20}) hits were generated by the internal oscillator in the FPGA to evaluate the linearity of the three types of TDCs.

To evaluate the precision of the TDC, two TDC channels employed with plain TDC, HU TDC, and HU-MCT TDC were implemented on the FPGA. The hit signals—which were square waves with a nearly full voltage swing (0.2–1.2 V) of the FPGA—were generated by an arbitrary function generator (AFG3102C, Tektronix) and sent to the two TDC channels. The frequency of the square wave was set to 10.18 MHz, which was uncorrelated to the system clock. In total, 65536 (2^{16}) timestamps were collected for each channel and stored in the internal memory of the FPGA. The stored timestamps were read by the PC through the UART interface, and the timestamp difference histogram between the two channels was acquired. The root mean square (RMS) precisions were calculated after applying a Gaussian fit to the timestamp difference histograms.

To evaluate the stability of the TDC against temperature fluctuations, the average bin widths and the RMS precisions as a function of temperature were measured for three types of TDCs. For this purpose, the operating temperature of the FPGA was adjusted from 34 °C to 74 °C in 10 °C steps by heating with a hair dryer, and the temperature value was monitored by the FPGA embedded thermometer.

3. Results

3.1. Linearity performance

Fig. 5 (a), (b), (c), and (d) show the relative and absolute DNLs/INLs acquired with the plain TDC, respectively. The ranges of relative and absolute DNLs/INLs were $[-1.0, 5.5]/[-24.0, 8.2]$ LSB and $[-1.9, 10.3]/[-45.2, 15.3]$ ps, respectively. The average bin width was calculated by taking the ratio of the system clock (2500 ps) to the number (1331) of effective bins covering the clock period; it was approximately 1.9 ps ($=1$ LSB).

Fig. 6 (a), (b), (c), and (d) show the relative and absolute DNLs/INLs acquired using the HU TDC, respectively. Compared to the plain TDC, the relative DNL and INL were significantly improved from $[-1.0, 5.5]$ to $[-0.9, 1.9]$ LSB and from $[-24.0, 8.2]$ to $[-6.4, 2.2]$ LSB, respectively, by using the HU TDC. Moreover, the absolute DNL and INL were improved from $[-1.9, 10.3]$ to $[-3.9, 7.9]$ ps and from $[-45.2, 15.3]$ to $[-26.7, 9.3]$ ps, respectively. However, the average bin width deteriorated from 1.9 to 4.2 ps ($=1$ LSB) because the number of effective bins covering the clock period was reduced from 1331 to 600 when using the HU method in TDC.

As shown in Fig. 7, the relative and absolute DNLs/INLs that were acquired using HU-MCT TDCs with 2, 3, and 4-chain TDLs (HU-2-MCT TDC, HU-3-MCT TDC, and HU-4-MCT TDC) were superior to those acquired using the HU TDC. The relative and absolute DNL/

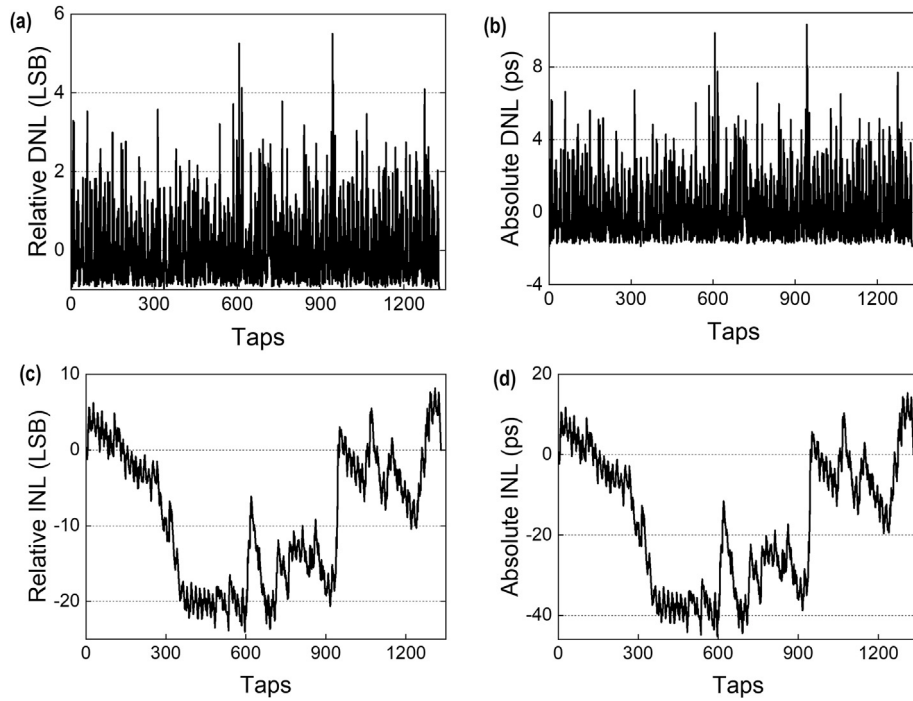


Fig. 5. Relative and absolute DNLs ((a), (b))/INLs ((c), (d)) acquired with the plain TDC (1 LSB = 1.9 ps).

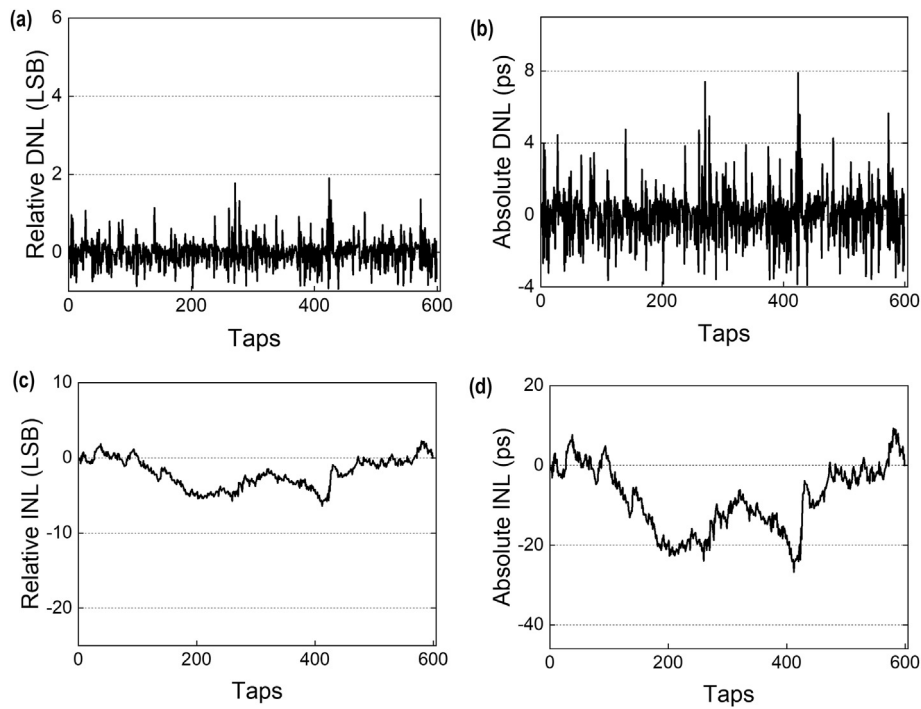


Fig. 6. Relative and absolute DNLs ((a), (b))/INLs ((c), (d)) acquired with the HU TDC (1 LSB = 4.2 ps).

INL were $[-0.8, 0.5]/[-1.7, 2.2]$ LSB and $[-3.3, 2.0]/[-7.1, 9.2]$ ps for HU-2-MCT TDC, $[-0.6, 0.2]/[-0.2, 2.3]$ LSB and $[-2.7, 0.8]/[-0.7, 9.8]$ ps for HU-3-MCT TDC, and $[-0.6, 0.1]/[-1.0, 1.2]$ LSB and $[-2.6, 0.4]/[-4.3, 5.1]$ ps for HU-4-MCT TDC. The average bin width was 4.2 ps (≈ 1 LSB) for all HU-MCT TDCs.

Table 1 summarizes the average bin widths as well as the

relative and absolute DNLs/INLs acquired with the plain TDC, HU TDC, HU-2-MCT TDC, HU-3-MCT TDC, and HU-4-MCT TDC.

3.2. RMS precision

The precision of the TDC was measured using two TDC channels

employed with plain TDC, HU TDC, and HU-3-MCT TDC. HU-3-MCT TDC was chosen among various HU-MCT-TDCs due to its excellent DNL and INL. Fig. 8 shows the RMS precisions that were obtained using the plain TDC, HU TDC, and HU-3-MCT TDC at different operating temperatures (34–74 °C) of the FPGA, and the timestamp difference histograms measured at specific operating temperatures (34 °C). The mean value of the timestamp difference was about 225 ps for all three types of TDCs. Compared to the plain TDC, the average RMS precision was improved from 7.3 ± 0.1 ps to 6.5 ± 0.1 ps by using the HU TDC, and it was further improved to 5.9 ± 0.1 ps by using HU-3-MCT TDC over the temperature range from 34 to 74 °C. The average variations in RMS precision over the entire temperature range were approximately 4.3% for the plain TDC, 4.6% for the HU TDC, and 4.4% for the HU-3-MCT TDC.

Fig. 9 shows the average bin widths as a function of the operating temperature of FPGA. The average bin widths were increased by 3.2% for the plain TDC, 3.3% for the HU TDC, and 3.6% for the HU-3-MCT TDC while increasing the operating temperature from 34 °C to 74 °C. The rates of increase of the average bin width against

temperature were approximately 0.0016 ps/°C for the plain TDC, 0.0035 ps/°C for the HU TDC, and 0.0037 ps/°C for the HU-3-MCT TDC.

4. Discussion

Newly introduced FPGAs have smaller delay elements and faster carry chains than the previous FPGAs, and they have allowed for the design of the TDL-based TDCs with a high time resolution. However, TDL-based TDCs suffer from bubble error and high nonlinearity caused by the poor uniformity of propagation delays in the TDL. In this study, to overcome such problems, TDL-based TDCs with the decimated delay line architecture (plain TDC), plain TDC with the histogram uniformization method (HU TDC), and HU TDC with the multi-chain TDL architecture (HU-MCT TDC) were proposed and implemented on the UltraScale + FPGA; the INLs, DNLs, and RMS precisions for the three types of TDCs were evaluated and compared to each other.

In the plain TDC, after applying the DDL architecture, bins up to

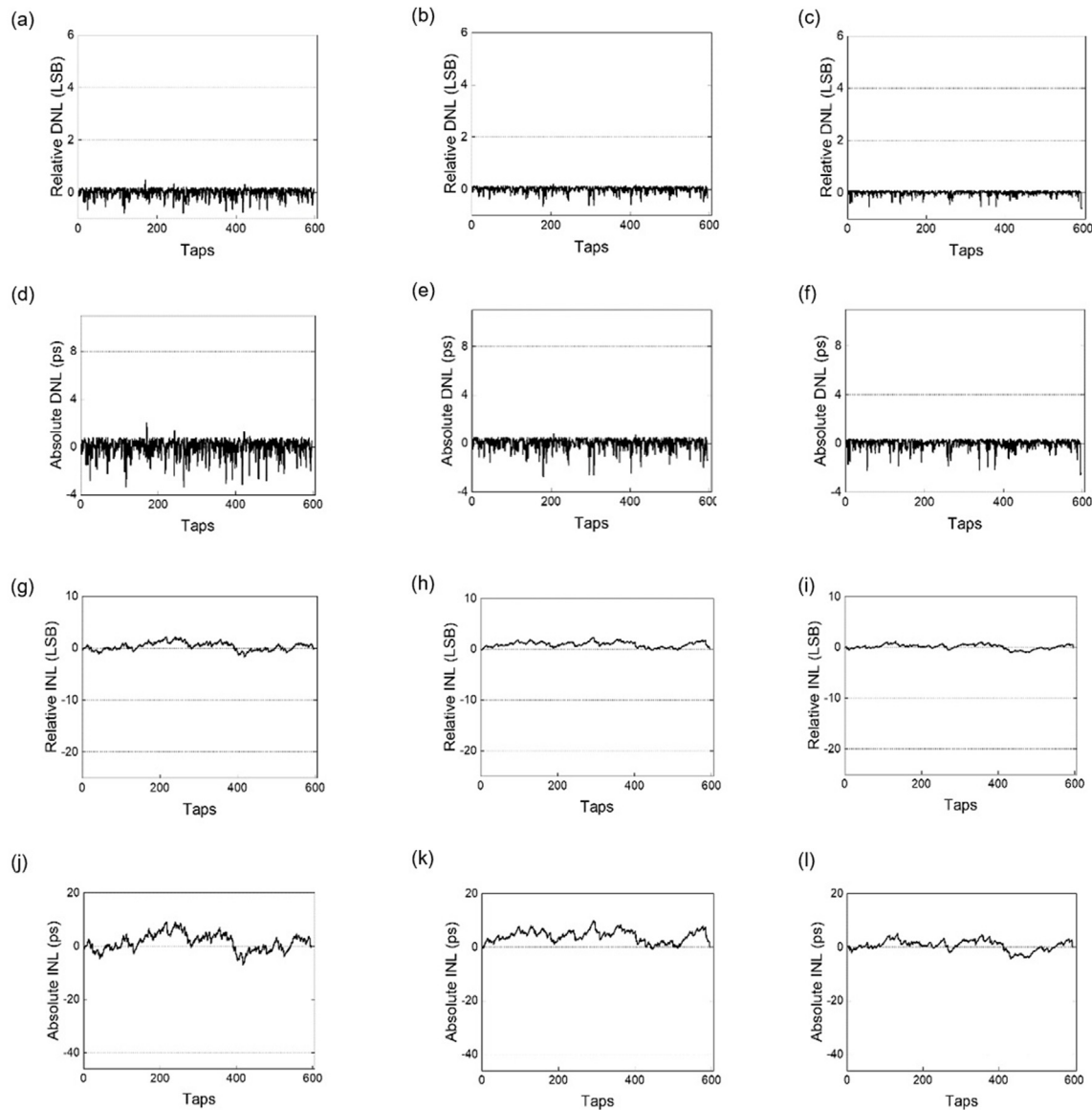
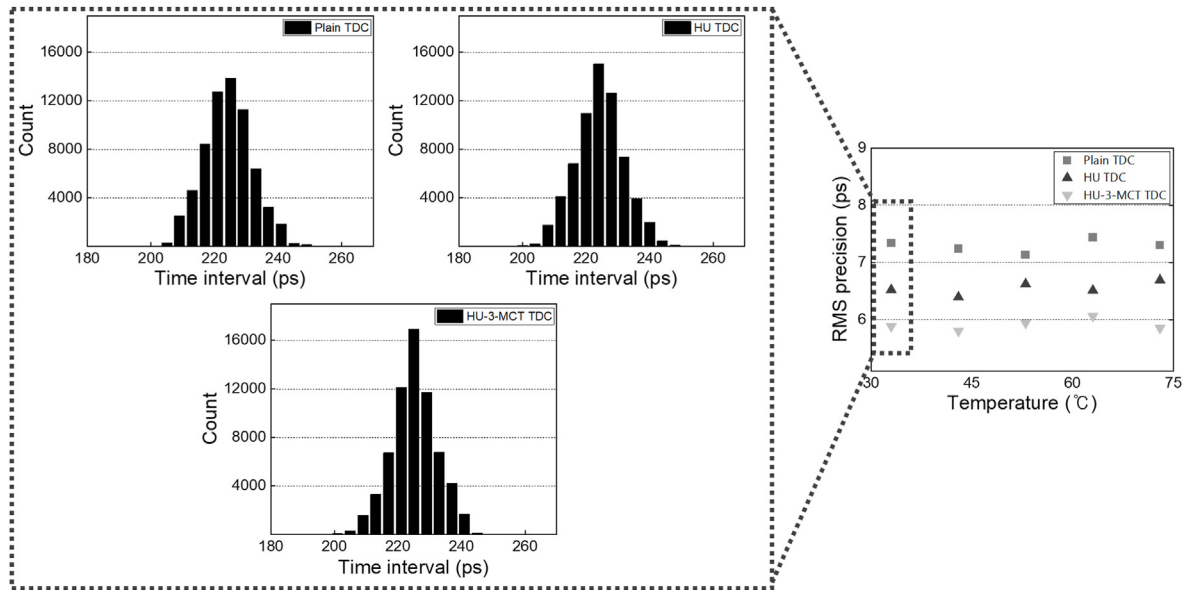


Fig. 7. Relative DNLs ((a), (b), (c))/INLs ((g), (h), (i)) and absolute DNLs ((d), (e), (f))/INLs ((j), (k), (l)) obtained with the HU-2-MCT TDC (left column), HU-3-MCT TDC (middle column), and HU-4-MCT TDC (right column) (1 LSB = 4.2 ps).

Table 1

Average bin widths and relative and absolute DNLs/INLs obtained with the plain TDC, HU TDC, HU-2-MCT TDC, HU-3-MCT TDC, and HU-4-MCT TDC.

	Average bin width (ps)	Relative DNL/INL (LSB)	Absolute DNL/INL (ps)
Plain TDC	1.9	[−1.0, 5.5]/[−24.0, 8.2]	[−1.9, 10.3]/[−45.2, 15.3]
HU TDC	4.2	[−0.9, 1.9]/[−6.4, 2.2]	[−3.9, 7.9]/[−26.7, 9.3]
HU-2-MCT TDC	4.2	[−0.8, 0.5]/[−1.7, 2.2]	[−3.3, 2.0]/[−7.1, 9.2]
HU-3-MCT TDC	4.2	[−0.6, 0.2]/[−0.2, 2.3]	[−2.7, 0.8]/[−0.7, 9.8]
HU-4-MCT TDC	4.2	[−0.6, 0.1]/[−1.0, 1.2]	[−2.6, 0.4]/[−4.3, 5.1]

**Fig. 8.** RMS precisions measured at different operating temperatures (34–74 °C) of the FPGA and representative timestamp difference histograms.

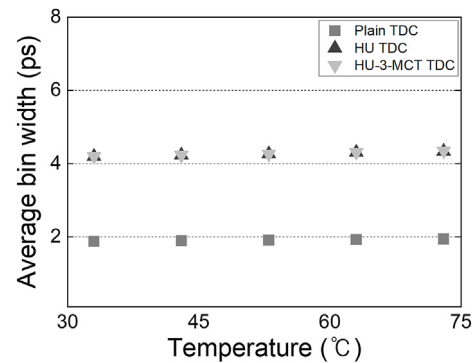
seven times wider than the average width and zero-width bins generated by the bubble error were both found to remain; there were three zero-width bins out of a total of 1331 effective bins. The minimum value of INL was very small (−24) because a cluster of bins having much smaller widths than its average was generated by the wide bins.

As shown in Figs. 5 and 6, the relative DNLs/INLs obtained with HU TDC were superior to those obtained with the plain TDC. Specifically, the minimum value of the relative DNL was −0.9, which indicates that the remaining zero-width bins were completely removed by the bin-merging process in the HU method.

To improve the precision of the TDC and reduce the absolute DNL and INL, the multi-chain TDL was employed in HU TDC. As shown in Fig. 7, the absolute DNLs/INLs were gradually improved by increasing the number of chains. However, the performance improvement was negligibly small when using more than 3-chains.

Although employing the multi-chain TDL in the TDC did improve the linearity, it can also cause a further increase in logic resources and power consumption. Therefore, the number of chains should be carefully determined while considering the tradeoff between linearity performance and resource utilization. To analyze the resource utilization of the three types of TDCs implemented on our FPGA, we checked the post-implementation utilization report provided by VIVADO 2018.2. As summarized in Table 2, the increase in resource utilization was negligibly small by applying the HU TDC to the plain TDC. However, using HU-3-MCT TDC led to an increase in the resource utilization by a factor of about three times compared to plain TDC or HU TDC.

As shown in Figs. 8 and 9, the RMS precision of HU-3-MCT TDC was superior to those of the plain TDC and HU TDC, which was

**Fig. 9.** Average bin widths measured at different operating temperatures (34–74 °C) of the FPGA.**Table 2**

Resource utilization of the plain TDC, HU TDC, and HU-3-MCT TDC implemented on our FPGA.

	Available	Plain TDC	HU TDC	HU3-MCT TDC
Carry8	27120	169 (0.62%)	172 (0.63%)	458 (1.69%)
LUT	216960	2642 (1.22%)	2710 (1.25%)	7186 (3.31%)
FF	433920	3187 (0.73%)	3442 (0.79%)	9411 (2.17%)
BRAM	480	4.5 (0.94%)	5 (1.04%)	17.5 (3.65%)
DSP	1824	1 (0.05%)	1 (0.05%)	1 (0.05%)

Table 3

Comparison of the proposed TDC with other TDCs.

	This work	Ref. [15]	Ref. [11]	Ref. [13]	Ref. [20]
Principle	TDL	TDL	TDL	TDL	TDL
FPGA vendor	Xilinx	Xilinx	Xilinx	Altera	Xilinx
Process technology (nm)	16	20	28	28	40
Number of bins	600	862	300	691	128
Reference clock (MHz)	400	500	200	250	600
RMS precision (ps)	5.9	4.2	8.1	6.6	12.8
Relative DNL (LSB)	[−0.6, 0.2]	—	[−1.0, 1.5]	[−1.0, 3.3]	[−1.0, 1.9]
Relative INL (LSB)	[−0.2, 2.3]	[−1.6, 2.0]	[−1.2, 4.3]	[−10.9, 11.7]	[−2.2, 2.6]

attributed to the fact that it had lower DNL and INL. As represented in Table 3, the HU-3-MCT TDC offered better RMS precision and DNL/INL values compared to other TDCs, except for RMS precision reported by Wang and Li [15]. However, the average bin width was increased by using HU TDC or HU-MCT TDC because several bins were merged by the HU method employed in TDC to improve the uniformity of the bin widths. Although the wide bins can degrade timing performance, this is not a serious problem in current commercial TOF-PETs, which have coincidence time resolutions ranging from 214 ps to 382 ps [30]. The variations in the RMS precisions and the average bin widths measured at different operating temperatures were both found to be negligibly small for all three types of TDCs, which was attributed to the fact that the 16 nm processing technology used for UltraScale + FPGA demonstrated excellent temperature stability [15].

5. Conclusion

In this study, an advanced TDC architecture capable of minimizing the bubble errors and improving the linearity was developed by implementing the decimated delay line architecture (plain TDC) and the histogram uniformization method (HU TDC) with the multi-chain TDL architecture (HU-MCT TDC) in the UltraScale + FPGA. The linearity of HU-MCT TDC were superior to those of the plain TDC and HU TDC, and it was further improved by increasing the number of chains of TDL in the HU-MCT TDC. The RMS precisions of all three types of TDCs remained stable over the temperature range from 34 °C to 74 °C. The experimental results indicated that the HU-MCT TDC developed in this study was useful for improving the linearity of the TDC, thus allowing for a high timing resolution to be achieved.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by the Korea Medical Device Development Fund grant funded by the Korea government (the Ministry of Science and ICT, the Ministry of Trade, Industry and Energy, the Ministry of Health & Welfare, the Ministry of Food and Drug Safety) (No. RS-2020-KD000006/1711137869 and RS-2020-KD000017/1711137911) and by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2019R111A1A01051112).

References

- [1] B.K. Swann, B. Blalock, L.G. Clonts, D. Binkley, J.M. Rochelle, E. Breeding, K.M. Baldwin, A 100-ps time-resolution CMOS time-to-digital converter for

positron emission tomography imaging applications, *IEEE J. Solid State Circ.* 39 (2004) 1839–1852.

- [2] A.S. Yousif, J.W. Haslett, A fine resolution TDC architecture for next generation PET imaging, *IEEE Trans. Nucl. Sci.* 54 (2007) 1574–1582.
- [3] G. Sportelli, N. Belcarì, P. Guerra, F. Spinella, G. Franchi, F. Attanasi, S. Moehrs, V. Rosso, A. Santos, A. Del Guerra, Reprogrammable acquisition architecture for dedicated positron emission tomography, *IEEE Trans. Nucl. Sci.* 58 (2011) 695–702.
- [4] J.Y. Won, J.S. Lee, Highly integrated FPGA-only signal digitization method using single-ended memory interface input receivers for time-of-flight PET detectors, *IEEE Trans. Biomed. Circuits Syst.* 12 (2018) 1401–1409.
- [5] E. Venialgo, N. Lusardi, F. Garzetti, Angelo Geraci, S. Brunner, D. Schaart, E. Charbon, Toward a full-flexible and fast-prototyping TOFPET block detector based on TDC-on-FPGA, *IEEE Trans. Radiat. Plasma Med. Sci.* 3 (2019) 538–548.
- [6] D. Tyndall, B. Rae, Day-Uei Li, J. Arlt, A. Johnston, J. Richardson, R. Henderson, A high-throughput time-resolved mini-silicon photomultiplier with embedded fluorescence lifetime estimation in 0.13 μm CMOS, *IEEE Trans. Biomed. Circuits Syst.* 6 (2012) 562–570.
- [7] H. Wang, Y. Yang, Z. Huang, H. Gui, Instrument for real-time measurement of low turbidity by using time-correlated single photon counting technique, *IEEE Trans. Instrum. Meas.* 64 (2015) 1075–1083.
- [8] K. Määttä, J. Kostamovaara, A high-precision time-to-digital converter for pulsed time-of-flight laser radar applications, *IEEE Trans. Instrum. Meas.* 47 (1998) 521–536.
- [9] J.P. Janssens, V. Koskinen, A. Mantyniemi, J. Kostamovaara, A multichannel high-precision CMOS time-to-digital converter for laser-scanner-based perception systems, *IEEE Trans. Instrum. Meas.* 61 (2012) 2581–2590.
- [10] C. Liu, Y. Wang, A 128-channel, 710 M samples/second, and less than 10 ps RMS resolution time-to-digital converter implemented in a kintex-7 FPGA, *IEEE Trans. Nucl. Sci.* 62 (2015) 773–783.
- [11] J.Y. Won, J.S. Lee, Time-to-digital converter using a tuned-delay line evaluated in 28-, 40-, and 45-nm FPGAs, *IEEE Trans. Instrum. Meas.* 65 (2016) 1678–1689.
- [12] J. Zheng, P. Cao, D. Jiang, Q. An, Low-cost FPGA TDC with high resolution and density, *IEEE Trans. Nucl. Sci.* 64 (2017) 1401–1408.
- [13] H. Xia, G. Cao, N. Dong, A 6.6 ps RMS resolution time-to-digital converter using interleaved sampling method in a 28 nm FPGA, *Rev. Sci. Instrum.* 90 (2019).
- [14] Y. Wang, C. Liu, A 3.9 ps time-interval RMS precision time-to-digital converter using a dual-sampling method in an UltraScale FPGA, *IEEE Trans. Nucl. Sci.* 63 (2016) 2617–2621.
- [15] Y. Wang, C. Liu, A 4.2 ps time-interval RMS resolution time-to-digital converter using a bin decimation method in an UltraScale FPGA, *IEEE Trans. Nucl. Sci.* 63 (2016) 2632–2638.
- [16] N. Lusardi, F. Garzetti, N. Corna, R. De Marco, A. Geraci, Very high-performance 24-channels time-to-digital converter in Xilinx 20-nm kintex UltraScale FPGA, in: *IEEE Nuclear Science Symposium (NSS) and Medical Imaging Conference (MIC)*, Manchester, UK, October 26 – November 2, 2019.
- [17] Y. Wang, Q. Cao, A multi-chain merged tapped delay line for high precision time-to-digital converters in FPGAs, *IEEE Trans. Circuits Syst. II-Express Briefs* 65 (2018) 96–100.
- [18] M.W. Fishburn, L.H. Menninga, C. Favi, E. Charbon, A 19.6 ps, FPGA-based TDC with multiple channels for open source applications, *IEEE Trans. Nucl. Sci.* 60 (2013) 2203–2208.
- [19] J. Torres, A. Aguilar, R. García-Olcina, P.A. Martínez, J. Martos, J. Soret, J. Benlloch, P. Conde, A. González, F. Sánchez, Time-to-digital converter based on FPGA with multiple channel capability, *IEEE Trans. Nucl. Sci.* 61 (2014) 107–114.
- [20] J.Y. Won, S.I. Kwon, H.S. Yoon, G.B. Ko, J.-W. Son, J.S. Lee, Dual-phase tapped-delay-line time-to-digital converter with on-the-fly calibration implemented in 40 nm FPGA, *IEEE Trans. Biomed. Circuits Syst.* 10 (2016) 231–242.
- [21] J. Wu, Several key issues on implementing delay line based TDCs using FPGAs, *IEEE Trans. Nucl. Sci.* 57 (2010) 1543–1548.
- [22] E. Bayer, M. Traxler, A high-resolution (< 10 ps RMS) 48-channel time-to-digital converter (TDC) implemented in a field programmable gate array (FPGA), *IEEE Trans. Nucl. Sci.* 58 (2011) 1547–1552.
- [23] X. Qin, L. Wang, D. Liu, Y. Zhao, X. Rong, J. Du, A 1.15-ps bin size and 3.5-ps single-shot precision time-to-digital converter with on-board offset

- correction in an FPGA, *IEEE Trans. Nucl. Sci.* 64 (2017) 2951–2957.
- [24] C. Hervé, J. Cerrai, T. Le Caër, High resolution time-to-digital converter (TDC) implemented in field programmable gate array (FPGA) with compensated process voltage and temperature (PVT) variations, *Nucl. Instrum. Methods Phys. Res.* 682 (2012) 16–25.
- [25] F. Garzetti, N. Corna, N. Lusardi, A. Geraci, Time-to-digital converter IP-core for FPGA at state of the art, *IEEE Access* 9 (2021) 85515–85528.
- [26] Y. Wang, J. Kuang, C. Liu, Q. Cao, A 3.9-ps RMS precision time-to-digital converter using ones-counter encoding scheme in a kintex-7 FPGA, *IEEE Trans. Nucl. Sci.* 64 (2017) 2713–2718.
- [27] M.-A. Daigneault, J.P. David, A novel 10 ps resolution TDC architecture implemented in a 130nm process FPGA, in: *IEEE International NEWCAS Conference*, Montreal, QC, Canada, June 20–23, 2010.
- [28] D. Chaberski, Time-to-digital-converter based on multiple-tapped-delay-line, *Measurement* 89 (2016) 87–96.
- [29] N. Lusardi, F. Garzetti, A. Geraci, The role of sub-interpolation for delay-line time-to-digital converters in FPGA devices, *Nucl. Instrum. Methods Phys. Res.* 916 (2019) 204–214.
- [30] S. Surti, J.S. Karp, Update on latest advances in time-of-flight PET, *Phys. Med.* 80 (2020) 251–258.